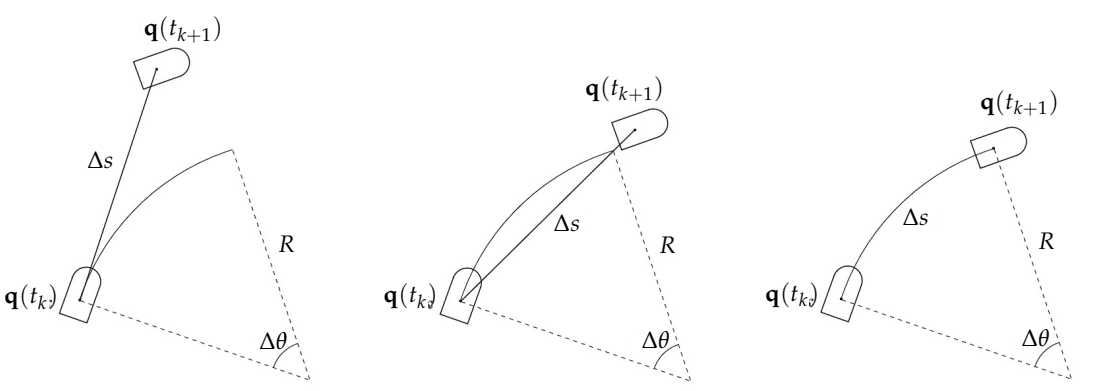

$\mathbf{q}(t_{k+1})$
Δs
R
$\mathbf{q}(t_k)$
$\Delta\theta$
Euler
$\mathbf{q}(t_{k+1})$
Δs
R
$\mathbf{q}(t_k)$
$\Delta\theta$
Runge-Kutta
$\mathbf{q}(t_{k+1})$
Δs
R
$\mathbf{q}(t_k)$
$\Delta\theta$
Exact